

SPYRIDON TSILIMPOKOS

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PROFILE

I build end-to-end systems across the full stack: from bare-metal drone firmware in C++ and self-supervised LSTM models to ML pipelines, AI agent architectures and production full-stack platforms. Recently completed MSc in Data Science & AI (Harokopio University). Looking for the right team to build something meaningful.

TECHNICAL SKILLS

Languages: Python, Java, R, SQL, JavaScript, C, Shell
AI / ML: PyTorch, Scikit-learn, LangGraph, HuggingFace, Qdrant, Spark, KNIME, Neo4j
Frameworks: FastAPI, Flask, Vue.js, Spring REST
Infra / Dev: Docker, MinIO, RabbitMQ, PostgreSQL, nginx, Git

PROFESSIONAL EXPERIENCE

Graduate Research Associate - Harokopio University Sept 2025 - Present
in collaboration with Hellenic Military Academy (SSE)

- Developing AI_Recovery: connectivity prediction & path recovery in GPS-denied SD-UAV networks (MSc thesis)
- Poster accepted & presented - Hellenic Robotics Forum 2026 (#0083)
- Paper submitted to IEEE Vehicular Technology Conference (IEEE VTC) - under review
- Paper in preparation - IARIA: 2nd Int. Conf. on AI-enabled UAVs & IoT for Critical Services

Independent Software Developer - Personal Projects 2022 - Present

- Project Atlas: full quadcopter build with custom C++ firmware (Madgwick AHRS, cascade PID, DShot600 via DMA); Android telemetry app in Godot 4 with full MAVLink v2 FSM
- SAR-UGV: multi-agent search & rescue on live OSM graphs, risk-weighted A*, 4 agents, 92% mission success across 50 scenarios
- AudioWeb: self-hosted YouTube to MP3 platform, WebGL audio visualizer, 10-band EQ, full Docker stack

Accounting & Administrative Support - Psimadas Marinos Nikolaos - Agrinio Jan 2019 - Dec 2021

- Handled daily accounting transactions, document filing and financial records management
- Provided administrative and secretarial support in a professional office environment
- Developed organisational skills and experience with client-facing financial communication

EDUCATION

MSc Data Science & AI - Harokopio University Oct 2025 - Sep 2026
G. Karampatzos Scholarship

BSc Informatics & Telematics - Harokopio University Sep 2021 - Sep 2025

RESEARCH & PUBLICATIONS

Poster accepted & presented - Hellenic Robotics Forum 2026 (#0083) 2026

SAR-UGV: Web-Based Multi-Agent Search & Rescue on Real Street Networks

Paper submitted - IEEE Vehicular Technology Conference (IEEE VTC) under review

Paper in preparation - IARIA: 2nd Int. Conf. on AI-enabled UAVs & IoT for Critical Services

MSc Thesis (in progress): AI_Recovery - Connectivity Prediction & Path Recovery in SD-UAV Networks

SELECTED PROJECTS

SAR-UGV Platform	FastAPI Python Docker A* Pathfinding NetworkX Leaflet.js OpenStreetMap
AudioWeb	Flask MinIO (S3) PostgreSQL Docker yt-dlp Web Audio API Butterchurn
Agentic RAG	Python LangGraph Chainlit Qdrant Docker
Computer Vision - U-Net/viT	PyTorch OpenCV U-Net CNN ViT SSIM
Food-11 Classification	PyTorch CNN ResNet50 VGG16 DenseNet121 Transfer Learning
MLP from Scratch	Python NumPy Backpropagation BCE Loss Skip Connections
Attention & Transformers	PyTorch HuggingFace BERT Self-Attention Cross-Attention
Graph Classification	PyTorch Geometric GCN GIN Node2Vec
NLP - Reddit Clustering	Python TF-IDF LSA Word2Vec Sentence Transformers HDBSCAN

LANGUAGES

English Michigan Proficiency - Fluent **German** Zertifikat B1 - Intermediate

CERTIFICATIONS & TRAINING

2025 - 2026 Design & Development of Unmanned Aerial Systems (UAS) - NKUA

2025 - 2026 Entrepreneurship & Strategic Planning for the Future - NKUA

2024 - 2025 Professional Skills - Harokopio University Career Liaison Office

2017 ECDL Advanced / Expert